MCQ 1: Understand and Choose

Five teaching questions with one answer. Choose one answer per question; then explain one This chapter format does not reproduce the official format of the EAM.

1. A distribution of probability has a sum of probabilities equal to

   A. 0
   B. 1
   C. n
   D. the average

2. X is 0 and 4 with probabilities $\frac{3}{4}$ and $\frac{1}{4}$; $E(X) =$

   A. 1
   B. 2
   C. 3
   D. 4

3. $V(X)$ is

   A. $E(X^2)$
   B. $E(X)^2$
   C. $E(X^2) + E(X)^2$
   D. $E(X^2) - E(X)^2$

4. If $V(X) = 9$, $\sigma(X) =$

   A. 81
   B. 9
   C. 3
   D. $\frac{1}{3}$

5. $E(2X + 1)$ is

   A. $2E(X)$
   B. $2E(X) + 1$
   C. $E(X) + 1$
   D. $E(X)^2 + 1$

MCQ 2: understand and choose

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1. $V(X + 5)$ is

   A. $V(X) + 5$
   B. $V(X) + 25$
   C. $V(X)$
   D. $5V(X)$

2. $\sigma(-3X)$ is

   A. $3\sigma(X)$
   B. $-3\sigma(X)$

C. $9\sigma(X)$

D. $\sigma(X)$

3. Expectation is necessarily

A. a possible value

B. One integer

C. Positive

D. None of these three statements

4. $E(X+Y)=E(X)+E(Y)$ requires

A. Independence

B. No independence

C. X=Y

D. Positive X

5. For $\sigma = 2$, n=100, $\frac{2\sigma}{\sqrt{n}} =$

A. 4

B. $0{,}04$

C. $0{,}4$

D. 2

True or false?

1. A variance may be negative.

2. Two variables of the same expectation have the same law.

3. Adding a constant to a variable does not change its standard deviation.

Personal review

I construct a complete law; I distinguish crude and clear; I control the two formulas of variance; I do not Confound not frequency and probability.
For each objective, indicate: acquired without help; succeeded with help; to rework. Write an example and a mistake to avoid.